LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can place rational numbers, including fractions and decimals, on a number line.

WHY THIS MATTERS Arihi planted a flower. The height and depth of the plant are shown. How do the height of the plant and the depth of the roots compare?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Rational number *Spanish: Número racional*	A number that can be written as a fraction of two integers, with a bottom number that is not zero.	−3/4, 0.5, 7, −2.85
Fraction *Spanish: Fracción*	A number that shows part of a whole, like 3/4.	3/4 of a cup
Decimal *Spanish: Decimal*	A number with a dot, like 0.5, that shows a part less than one.	0.5 is half of one
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.	left is less, right is more
Equivalent *Spanish: Equivalente*	Having the same value, just written a different way.	1/2 = 0.5
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Plot 1.5 and −3.5 on a number line.

1. Mark the integers from −4 to 4.
2. Split each gap into halves.
3. 1.5 is halfway between 1 and 2. −3.5 is halfway between −4 and −3.

Answer: 1.5 is right of 0; −3.5 is left of 0

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's plot 0.75. Which two integers is it between? It is between 0 and 1.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The number _____ is rational because it can be written as _____ .
- On the number line, _____ falls between _____ and _____ .
- I split each unit into _____ equal parts because _____ .
- The opposite of _____ is _____ , so it sits to the _____ of zero.

Word bank rational number | fraction | decimal | opposite | number line | integer | between | equal parts | denominator | above | below | zero

WATCH OUT!

A common mistake

A common mistake in Rational Numbers on the Number Line is treating the numerator and denominator of a negative fraction as two separate integers instead of one value between two integers

LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each fraction to its decimal equivalent.

_____ 1/2	(A) 0.5
_____ -3/4	(B) -0.75
_____ 1 1/4	(C) 1.25
_____ -2 1/2	(D) -2.5
_____ 3/10	(E) 0.3
_____ -1/5	(F) -0.2

HINT Each fraction equals a decimal. Halves, quarters, fifths, and tenths all have decimal partners you know.

2 **Mark and label each value on the number line.**

Plot these rational numbers: -1, 0.5, -0.5, 1, -1.5



HINT The tick marks count by 0.5 — each hop is a HALF, not a whole unit.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which point is located between -1 and -2 on the number line?

(A) -2.5
(B) -1.5
(C) -0.5
(D) 1.5

HINT 'Between -1 and -2' is on the NEGATIVE side, in the gap from -1 to -2.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which fraction is equivalent to 0.25?

(A) 1/4
(B) 1/2
(C) 1/3
(D) 2/5

HINT Read 0.25 as a place-value number: 25 hundredths.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Where is -3/4 located on the number line?

(A) Between -3 and -4
(B) Between -1 and -2
(C) Between 0 and -1
(D) Between 0 and 1

HINT The size of -3/4 is less than 1 whole — its distance from zero is under 1.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Mistake: Plotting 1.5

1. **Problem** Plot 1.5 on the number line.
2. **Student thinking** 1.5 has a 1 and a 5, so I will plot it between 1 and 5.
3. **Student answer** Point plotted at about 3, halfway between 1 and 5.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** What does the .5 in 1.5 mean — is it a separate whole number or a part of one?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide exactly where to place rational numbers that fall between integers?

> **SENTENCE STARTER** To plot ___, I found the integers it falls between, which are ___ and ___. Then I estimated ___ of the way between them because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Plot these rational numbers: -3.5, -1/4, 2.75, -2, 1 1/2



SHOW YOUR THINKING

2 **Complete the table. Show how you found each value.**

Convert between fraction and decimal form. Then plot on a mental number line to compare.

Fraction	Decimal	Between Which Two Integers?
3/4		
-7/2		
	0.6	
	-1.25	
2 1/5		

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Order these rational numbers from LEAST to GREATEST.

Groups:

_____ 1/2

_____ -2.5

_____ 0.75

_____ -1/4

_____ 3

_____ -1.5

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Which rational number is closest to 0 on the number line?

(A) -2

(B) -1/4

(C) 0.5

(D) 1.75

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which number is farthest to the LEFT on a number line?

(A) -4 (B) -1

(C) 0 (D) 2

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake: Plotting -2.25

1. **Problem** Plot -2.25 on the number line.
2. **Student thinking** It is negative, so I go past -2. The .25 is one quarter, so I count toward 0 (to the right) by a quarter unit and land between -2 and -1.
3. **Student answer** Point plotted at -1.75, between -2 and -1.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide exactly where to place rational numbers that fall between integers?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Decide whether each pair balances. Check Yes or No, then explain.**

Which side is greater? Compare -2/3 and -3/4.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

2 **Circle the letter of the best answer. Show how you know.**

On a number line, point P is located at -2.5. Which statement describes its location best?

(A) At -3

(B) At -2

(C) Halfway between -2 and -3

(D) Halfway between 2 and 3

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which list of numbers is ordered from LEAST to GREATEST?

(A) -5, -2, 0, 3

(B) -2, -5, 0, 3

(C) 0, -2, 3, -5

(D) 3, 0, -2, -5

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Find the mistake. Explain it, then show the correct work.**

Find the Plotting Error

1. **Problem** Plot -3/4 on the number line
2. **Student thinking** -3/4 has a 3 and a 4, so I go to -3 on the number line and then count 4 more to the left
3. **Student answer** Point plotted at -7

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

5 **Answer in complete sentences.**

Name three different rational numbers between -1 and 0. Write each as both a fraction and a decimal. Explain how you know they are between -1 and 0.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did you decide exactly where to place rational numbers that fall between integers?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help